

Name: _____



Features of geometric sequences

Task A: Identifying geometric sequences

1) Sort these sequences into **geometric**, arithmetic and 'other'.

a) 1, 2, 3, 4, 5, ...

g) 4, 16, 30, 46, 64, ...

b) 1, 2, 4, 8, 16, ...

h) 10, 15, 20, 30, 40, ...

c) 1, 2, 5, 10, 17, ...

i) 10, 40, 70, 100, 130, ...

d) -1, -2, -4, -8, -16, ...

j) 10, 20, 60, 240, 1200, ...

e) 1, 4, 16, 64, ...

k) 10, 20, 40, 80, 160, ...

f) 4, 16, 28, 40, 52, ...

l) 10, 100, 1000, 10 000, ...

Task B: Generating geometric sequences

1) Generate the first five terms of these **geometric sequences**.

- a) 1st term 1, **common ratio** of 3
- b) 1st term 10, **common ratio** of 3
- c) 1st term 5, ratio 1 : 3 between terms
- d) 1st term 3, a ratio 1 : 5 between terms
- e) 1st term 64, a ratio 2 : 5 between terms
- f) 1st term 81, **common ratio** of $\frac{1}{3}$
- g) 1st term 1875, a ratio 5 : 1 between terms
- h) 1st term 3, a ratio 3 : 1 between terms

2) Alex is generating these **geometric sequences**.

- a) 1st term 216, **common ratio** $\times \frac{1}{3}$
- b) 1st term 216, **common ratio** $\times \frac{2}{3}$
- c) 1st term 216, **common ratio** $\times \frac{3}{2}$



Because these multipliers are fractions,
all of these will be decreasing sequences.

Show Alex that he is wrong.

